UNITED UNIVERSITY

Faculty of Commerce & Management (BBA-IBM)

PREDICTIVE ANALYTICS

PRACTICAL PROJECT REPORT

Customer Churn Prediction Using IBM SPSS Modeler

A telecom case study using the Auto Classifier node

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Semester	5th
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Academic Session	2026-27

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Abstract

This practical project uses IBM SPSS Modeler to predict customer churn for a telecom company. Churn means that a customer stops using the company's service. If the company can identify such customers early, it can try to keep them with offers or better service.

The training file, telco x customer data.xlsx, has 31,781 customer records and 12 fields, including gender, age, region, connection date, dropped calls, tariff, handset and the target field churn (Active or Churned). A second file, telco_test_dataset.xlsx, has 20 new customers without a churn value. The stream was built as follows: Excel source node, Type node, Sort node, Auto Classifier node, a generated model nugget, and Table nodes to view the data and predictions.

The Auto Classifier node estimated 12 models and kept the best three, ranked by overall accuracy. In the model summary seen in the training-set view, C5 gave 89.903% accuracy, Neural Net 88.975% and Bayes Net 87.637%. The combined model was then applied to the 20 test customers: 14 were predicted as Churned and 6 as Active. Accuracy on the 20 test customers cannot be calculated because their real churn status is not available. The report states clearly which results are available from the project and which are not.

1. Introduction

1.1 Background

Mobile and telephone companies compete for the same customers. Winning a new customer usually costs much more than keeping an existing one, so telecom companies closely watch customer churn. A churned customer is one who has left the network. An active customer is one who is still using the service.

1.2 Predictive Analytics and Churn

Predictive analytics uses past data to estimate what is likely to happen in the future. In this project, the past data is a table of customers whose churn status is already known. A model learns the patterns in this table, such as which handsets, tariffs and call experiences are common among customers who left. The model is then used on new customers to estimate whether they are likely to stay or leave.

1.3 Why IBM SPSS Modeler

IBM SPSS Modeler is a visual data-mining tool. Instead of writing code, the user places small icons called nodes on a canvas and joins them with arrows. A connected set of nodes is called a stream. Each node does one job, such as reading a file, defining field types, sorting records, building a model or showing a table. This makes the tool suitable for business students who want to understand the full predictive process without programming.

1.4 Importance of this Practical

This practical shows a complete, small predictive-analytics workflow: reading data, preparing it, building models, checking model performance and predicting for new customers. It also links to the CRISP-DM stages shown in the tool (Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation and Deployment).

2. Problem Statement

A telecom company has records of 31,781 past customers. For each customer it knows personal and usage details and whether the customer is Active or has Churned. The company wants a way to predict the churn status of a customer from these details.

- **What is predicted:** The field churn, a two-value (flag) field with the values Active and Churned.
- **Why it is needed:** Customer retention costs money, and it is wasteful to give offers to everyone. A prediction helps the company focus on customers who are more likely to leave.
- **Who benefits:** The marketing, customer care and finance teams, who can plan retention offers, call-quality improvements and budgets.
- **Type of prediction:** Because the target has two possible outcomes, this is a binary classification problem, not a regression problem.

Problem in one line: *Using the customer details in the training file, build a model in IBM SPSS Modeler that predicts whether a customer will be Active or Churned, and use it on a new set of customers.*

3. Objectives of the Project

- To understand the structure and meaning of the telecom customer data.
- To build a working SPSS Modeler stream that reads the data from Excel files.
- To set the correct field types and mark churn as the target field.
- To use the Auto Classifier node to build and compare several models at the same time.

- To read the model results (accuracy, profit, lift and area under the curve) and the predictor importance chart.
- To apply the generated model to a new test file and read the predicted churn and confidence values.
- To explain the results in simple business language and be clear about what the results do not show.

Expected outcome: A model that gives each customer a predicted churn status with a confidence score, and a list of the fields that matter most for the prediction.

4. Software and Tools Used

The practical was done fully inside IBM SPSS Modeler. The stream file was saved from the program with the stream name Stream1. No Python code or libraries were used.

Tool	Use in this project
IBM SPSS Modeler (version 18 stream file)	Main tool. Used to build the stream, run the models and view predictions. The status bar shows it ran on the Local Server.
Microsoft Excel (.xlsx files)	Stores the two datasets: telco x customer data.xlsx (training data) and telco_test_dataset.xlsx (new customers to predict).
Auto Classifier node	Builds and compares many classification models in one run and combines the best ones.
Table, Type and Sort nodes	Used to view data, define field types and arrange records.
Windows computer	Operating system used to run the software.

Table 1: Tools and their purpose

5. Dataset Description

5.1 Files Used

Two Excel files were used. The training file contains the answer (churn) for every customer, so the model can learn from it. The test file does not contain the churn field, because it represents new customers whose result is not yet known.

Item	Training file	Test file	Remark
File name	telco x customer data.xlsx	telco_test_dataset.xlsx	Both read by Excel nodes
Records (rows)	31,781	20	Test file is a small sample
Fields (columns)	12	10	Test file has no end_date or churn
Target field	churn	Not present	To be predicted
Source	Provided for the practical	Provided for the practical	Public link not known

Table 2: Overview of the two datasets

5.2 Fields in the Training Data

Field	Kind	Meaning
customer_id	Text (ID)	Unique customer code such as K100010
gender	Text	Male or Female (written in several spellings in the file)

Field	Kind	Meaning
age	Number	Customer age in years
postalcode	Number	Postal code of the customer (1000 to 9998)
region	Number	Region code, 1 to 4
connect_date	Date	Date the customer joined (2003-01-01 to 2006-12-31)
end_date	Date	Date the connection ended; empty for customers who are still Active
dropped_calls	Number	Number of dropped calls (0 to 45)
pay method	Text (flag)	Payment method. Every record shows Pre Pay
tariff	Text	Plan type: CAT 50, CAT 100, CAT 200, Play 100, Play 300
handset	Text	Phone model, 12 types such as S50, BS110, ASAD90, WC95
churn	Text (flag) – Target	Active or Churned

Table 3: Field description (training file)

5.3 Target Variable Distribution

In the training file, 17,062 customers (53.7%) are Active and 14,719 customers (46.3%) are Churned. The two groups are fairly balanced, which is good for model building because the model sees many examples of both outcomes.

5.4 Data Quality Points Noticed

The following points were found by checking the Excel training file. They are important for understanding the results.

- **Gender spelling:** Gender appears in six spellings (Male, male, MALE, Female, female, FEMALE). The Type node therefore shows it as a set with six values. A cleaning step would merge them into two.
- **Duplicates:** About 12 repeated records exist (the same customer_id appears twice in 12 cases). No duplicate-removal node was used in the stream.
- **Age values:** The age field has a minimum of −1 and a maximum of 82. Fifteen records have an age of zero or less. These are unusual values.
- **end_date:** end_date is empty for all 17,062 Active customers and filled for 14,686 of the 14,719 Churned customers. So end_date is almost the same information as churn itself. The test file does not have this field, so a model that is applied to the test file cannot depend on it.
- **pay method:** Only one value (Pre Pay) exists, so this field cannot help to separate Active from Churned customers.
- **Date format in the test file:** In the test table the connect_date is shown as a number (for example 37861), which is how Excel stores dates internally, while the training file shows dates in year-month-day form.

6. Practical Procedure (Step by Step in IBM SPSS Modeler)

The steps below follow the actual stream. The stream has an Excel source for training data, a Type node, a Sort node, an Auto Classifier node, one generated model, a second Excel source for test data, and five Table nodes.

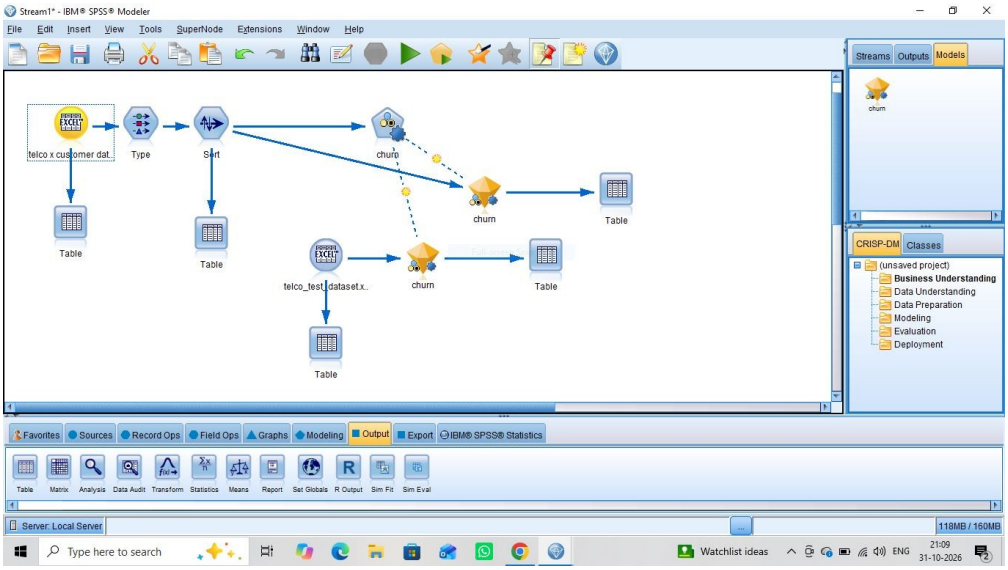


Figure 1: Complete stream (Stream1) in IBM SPSS Modeler

Step 1: Open IBM SPSS Modeler and start a new stream

Open the program. A blank stream canvas appears in the middle. The bottom palette holds the node groups (Sources, Record Ops, Field Ops, Graphs, Modeling, Output, Export). The panel on the right has the Streams, Outputs and Models tabs and the CRISP-DM project tree. The stream is named Stream1.

Step 2: Add the Excel source node for the training data

From the Sources tab, place an Excel node on the canvas and select the file telco x customer data.xlsx. The first row is used as field names. This node only reads data; it does not change it.

Step 3: Attach a Table node to the Excel node

From the Output tab, add a Table node and join it to the Excel node. Running it shows all 31,781 records and 12 fields in a spreadsheet-like window. This is the first look at the data (customer_id, gender, age, postalcode, region, connect_date, end_date, dropped_calls and so on).

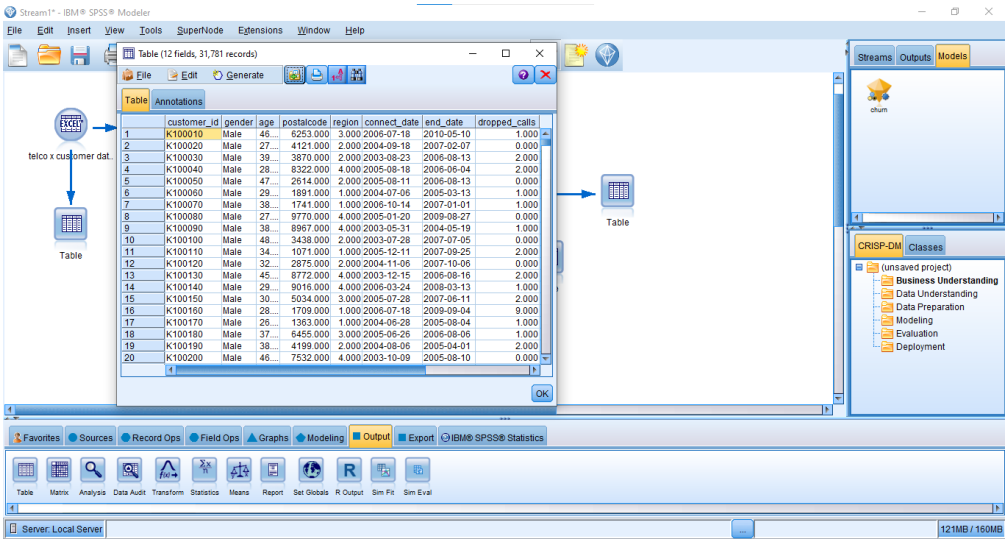


Figure 2: Table node showing the raw training data (31,781 records, 12 fields)

Step 4: Add and set the Type node

From the Field Ops tab, add a Type node after the Excel node. The Type node tells the software the measurement level and the role of every field. After clicking Read Values, the node shows the range or the list of values of each field. In this stream, numeric fields such as postalcode, region, dropped_calls and age are Continuous; connect_date and end_date are dates; pay method is a Flag; tariff and handset

are Nominal. The field churn (values Active and Churned) is set to the role Target, and the other visible fields have the role Input.

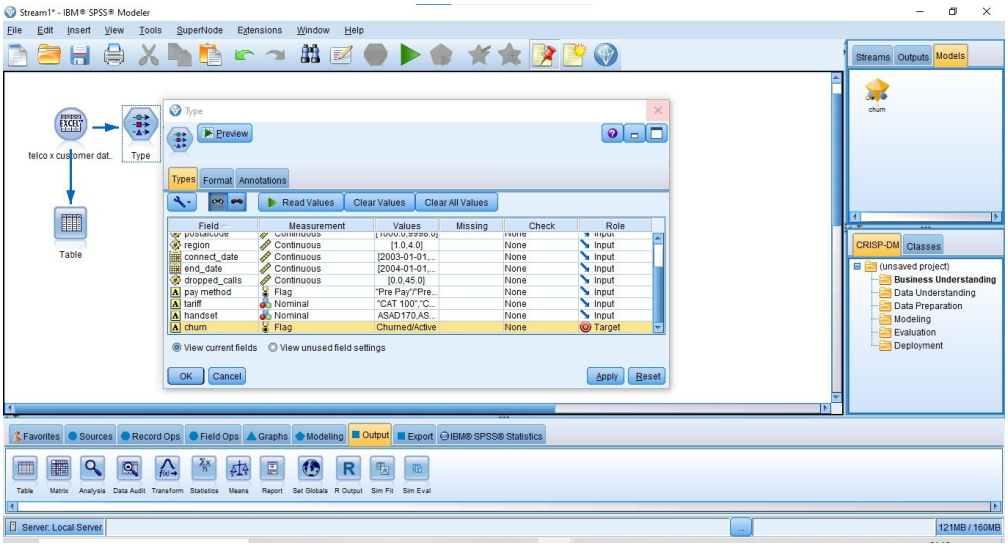


Figure 3: Type node: field types and roles, with churn set as Target

Step 5: Add the Sort node

From the Record Ops tab, add a Sort node after the Type node. The sort key is connect_date in Ascending order. After sorting, the first records are customers who connected on 2003-01-01. Sorting only changes the order of the records; it does not change any values, and it is not needed for the model to work. Here it arranges customers from the oldest to the newest connection.

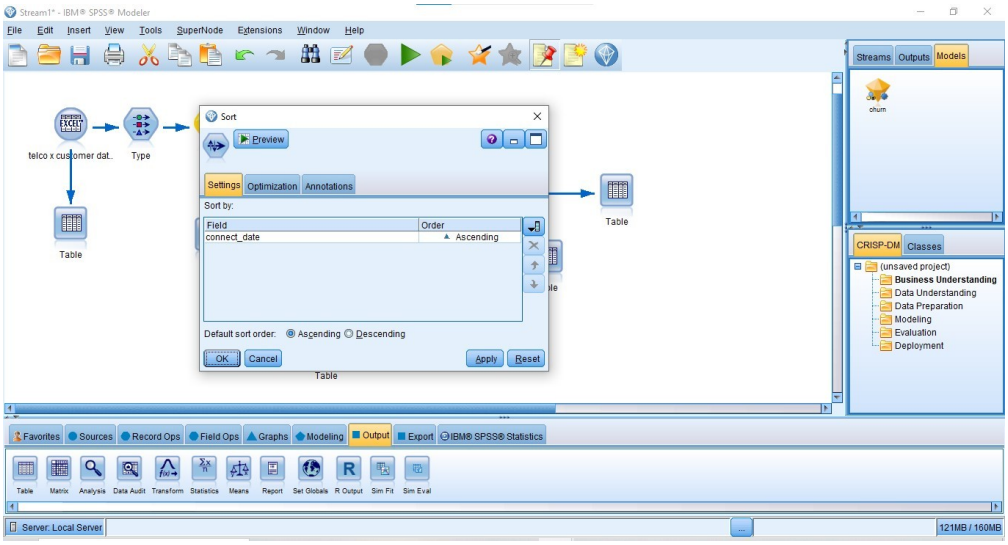


Figure 4: Sort node: connect_date in ascending order

Step 6: Attach a Table node to the Sort node

A second Table node is joined to the Sort node. It shows the 31,781 records in connect_date order. In this view, end_date is displayed as $null$ for customers who have not left. SPSS Modeler uses $null$ to show an empty value.

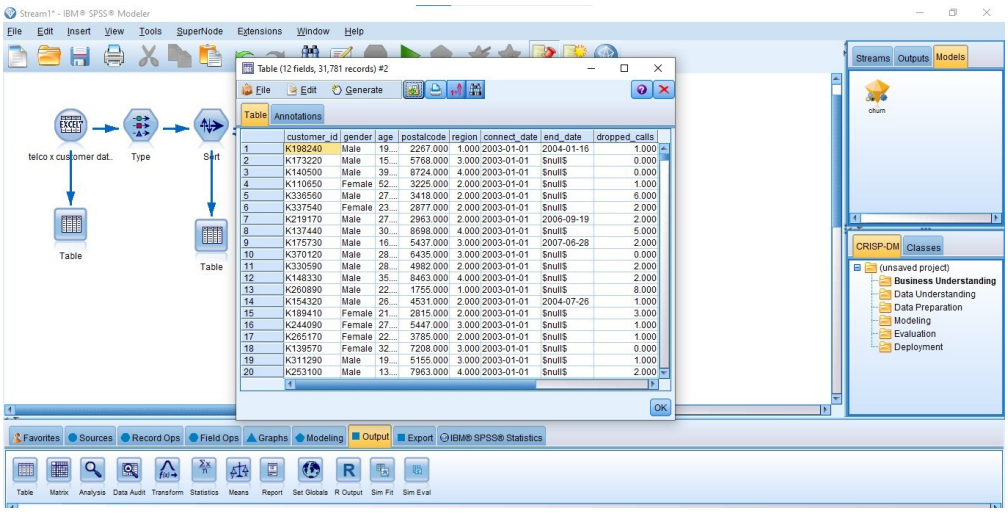


Figure 5: Table node after Sort (end_date shown as $null$ for Active customers)

Step 7: Add the Auto Classifier node and name it churn

From the Modeling tab, add an Auto Classifier node after the Sort node. Because churn is the Target, the node knows it must build classification models. Its settings are described in Section 7. The node window shows “Estimated number of models to be executed: 12”.

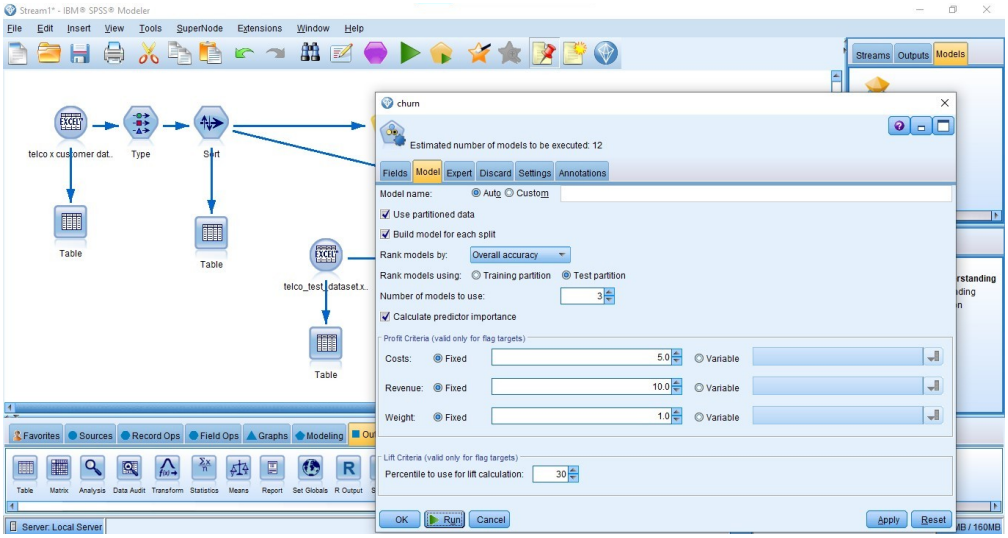


Figure 6: Auto Classifier node settings (Model tab)

Step 8: Run the Auto Classifier node

Click Run. The node builds the models, ranks them and places a golden model nugget named churn in the Models tab on the right. The nugget is also placed on the canvas and connected to the Sort node.

Step 9: Open the model nugget and study its tabs

Open the nugget. The Model tab lists the models that were kept with Max Profit, Lift, Overall Accuracy, Number of Fields Used and Area Under Curve. The Graph tab shows a Distribution chart and a Predictor Importance chart. The Summary, Settings and Annotations tabs give further details.

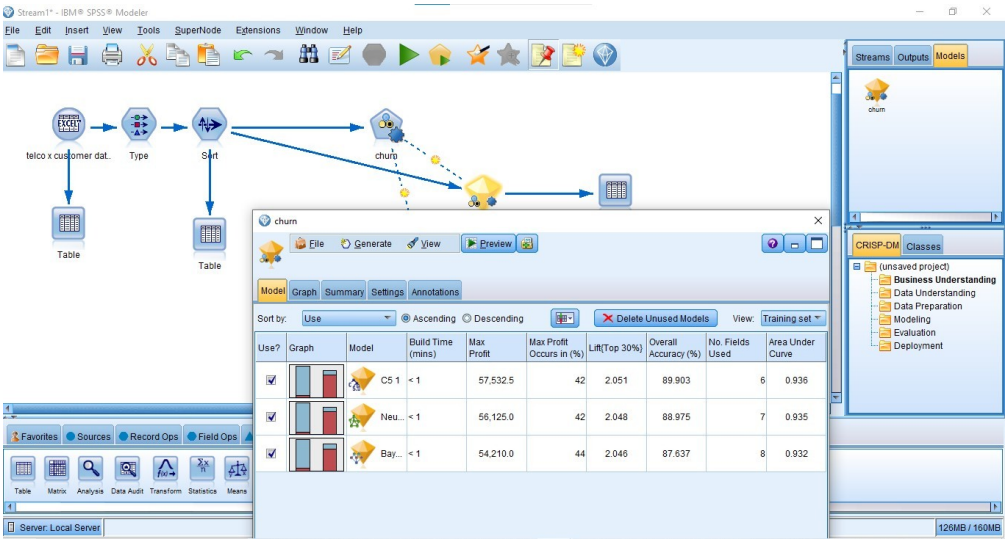


Figure 7: Model nugget: Model tab with the three models kept

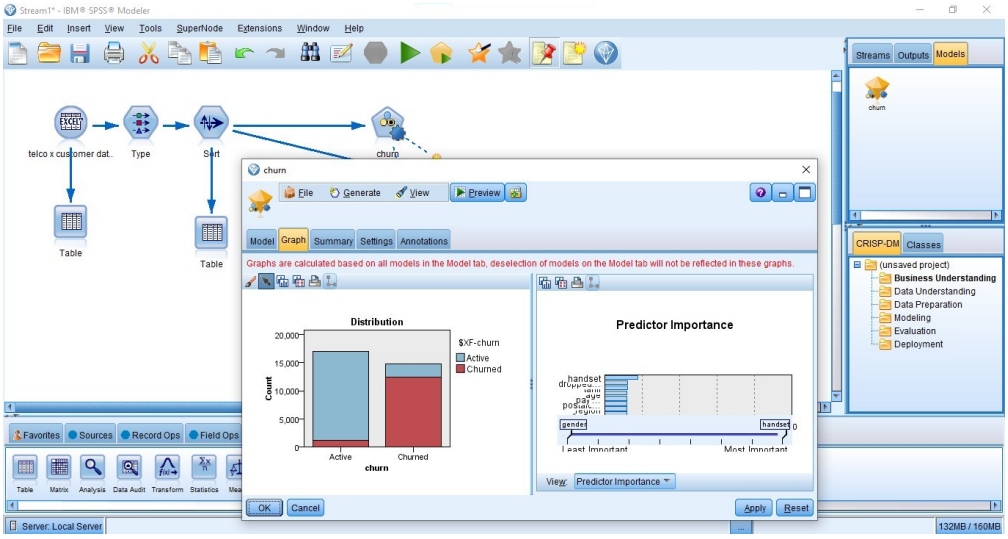


Figure 8: Model nugget: Graph tab (Distribution and Predictor Importance)

Step 10: Attach a Table node to the nugget (predictions on training data)

A Table node is joined to the nugget. Running it shows 31,781 records with 14 fields: the original 12 plus two new fields, $XF-churn (predicted churn) and $XFC-churn (confidence of the prediction).

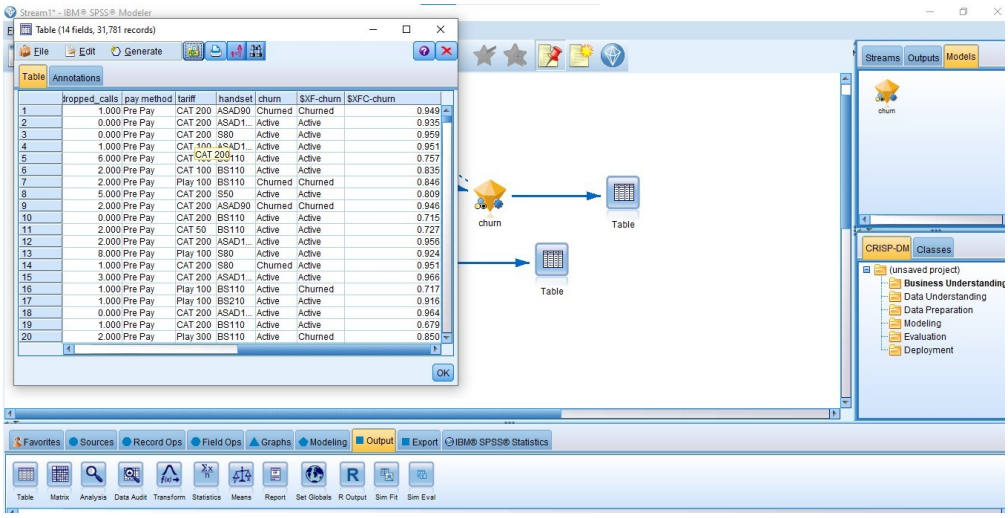


Figure 9: Table node after the nugget: $XF-churn and $XFC-churn added (training data)

Step 11: Add the second Excel source node for the test data

A second Excel node reads telco_test_dataset.xlsx. A Table node attached to it shows 20 records and 10 fields. There is no churn field here because these are new customers.

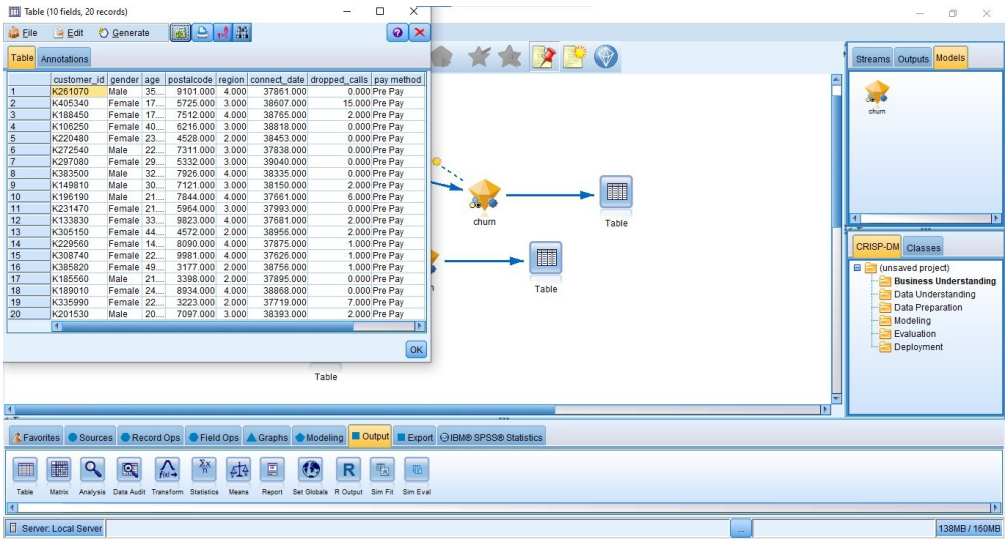


Figure 10: Test dataset table (20 records, 10 fields)

Step 12: Apply the model to the test data

The same churn model nugget is connected to the test Excel node and a Table node is attached after it. Running this table shows 20 records and 12 fields: the 10 test fields plus $XF-churn and $XFC-churn. These are the final predictions for the new customers.

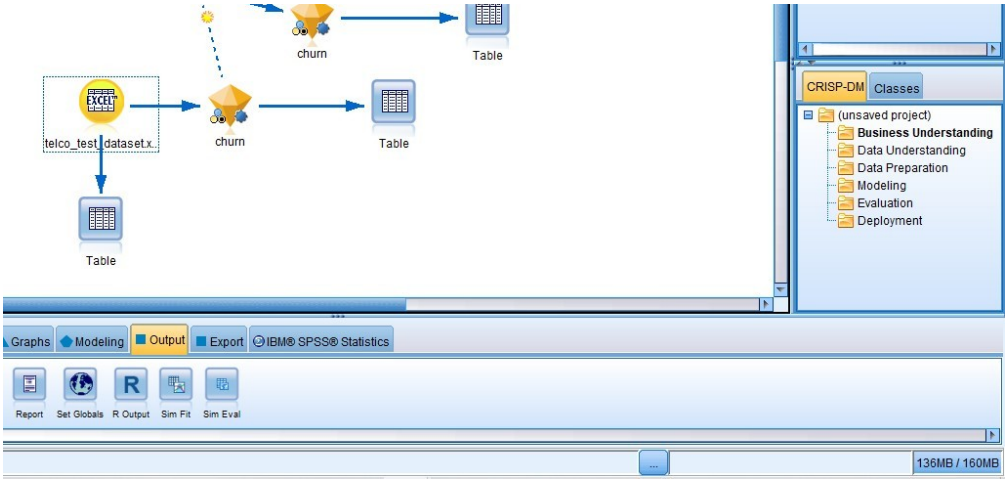


Figure 11: Test Excel node connected to the churn model nugget and a Table node

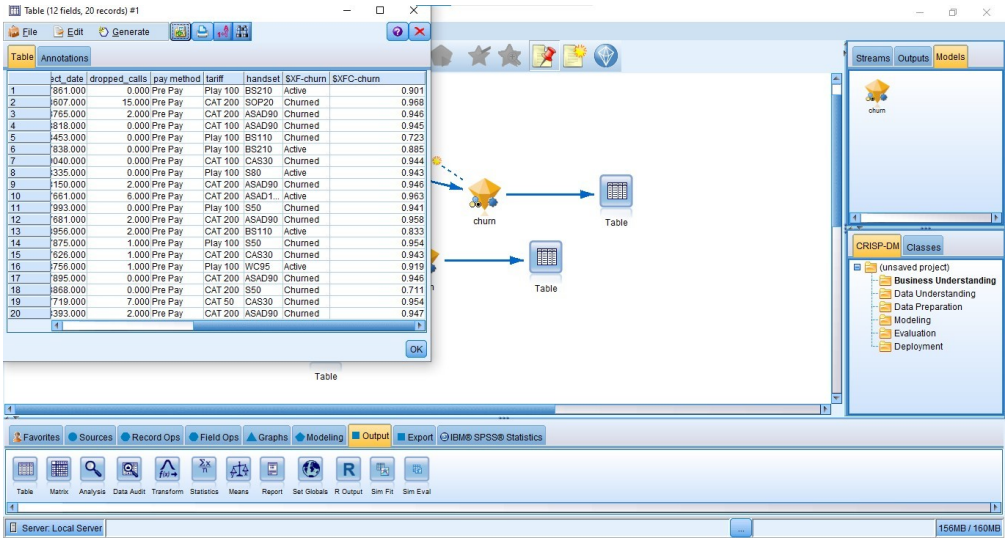


Figure 12: Predictions for the 20 test customers

6.1 Purpose of Each Node Used

Node	Palette	Purpose in this stream
Excel (training)	Sources	Reads telco x customer data.xlsx into the stream (31,781 records).
Table (after Excel)	Output	Shows the raw data for checking.
Type	Field Ops	Defines field types and roles; sets churn as Target and others as Input.
Sort	Record Ops	Orders records by connect_date from oldest to newest.
Table (after Sort)	Output	Shows the sorted data.
Auto Classifier (churn)	Modeling	Builds several classification models, ranks them and keeps the best three.
Model nugget (churn)	Models	Stores the final combined model and adds prediction fields to any data passed through it.
Table (after nugget, training)	Output	Shows actual churn next to predicted churn and confidence.
Excel (test)	Sources	Reads telco_test_dataset.xlsx (20 new customers).
Table (after Excel, test)	Output	Shows the test data before prediction.
Table (after nugget, test)	Output	Shows the predictions for the 20 new customers.

Table 4: Nodes in the stream and their purpose

7. Model Building and Prediction

7.1 Why the Auto Classifier Node

The target field churn has two values, so a classification model is needed. Different algorithms work better on different data. The Auto Classifier node builds many models in one run, compares them and keeps the best ones. This saves time and lets the student compare techniques without building each one by hand. The stream estimates 12 models in this run.

7.2 Settings Used in the Auto Classifier Node

Setting	Value in the stream
Target field	churn (Flag: Active / Churned)

Setting	Value in the stream
Use partitioned data	Ticked (but no Partition node is present in the stream)
Build model for each split	Ticked
Rank models by	Overall accuracy
Rank models using	Test partition
Number of models to use	3
Calculate predictor importance	Ticked
Profit criteria	Cost = 5.0, Revenue = 10.0, Weight = 1.0 (all fixed)
Lift criteria	Percentile for lift calculation = 30
Combining predictions (ensemble)	Confidence-weighted voting (shown by the output fields $XF-churn and $XFC-churn)

Table 5: Auto Classifier settings

7.3 Meaning of the Main Settings

- **Overall accuracy:** The share of all predictions that are correct. Models are ranked using this value.
- **Profit criteria:** The profit setting assumes a fixed cost of 5 and a fixed revenue of 10 per customer. The software uses these to estimate the profit if the best-scored customers are selected.
- **Lift:** Shows how many times better the model is than random choice when only the top 30% of scored customers are taken.
- **Area Under Curve (AUC):** A value between 0 and 1. Closer to 1 means the model separates Active and Churned customers better.
- **Predictor importance:** Ranks the input fields by how much they help the prediction.

7.4 Prediction Output

When data passes through the model nugget, two fields are added. $XF-churn is the predicted class (Active or Churned) and $XFC-churn is the confidence, a number between 0 and 1 showing how sure the combined model is about its prediction. For example, a confidence of 0.95 means the model is very sure, while 0.71 means it is less sure.

Important note on data split: the Model tab shows a “Training set” view, and the stream contains no Partition node. So the accuracy values reported in Section 8 are those shown in the training-set view of the model. A separate testing result for these 31,781 customers is not shown in the project, and it is therefore not reported here.

8. Observations and Results

8.1 Models Kept by the Auto Classifier

The Model tab of the nugget shows three models. The values below are taken from the training-set view. All three models took less than one minute to build.

Model	Max Profit	Max Profit at (%)	Lift (Top 30%)	Overall Accuracy (%)	Fields Used	AUC
C5	57,532.5	42	2.051	89.903	6	0.936
Neural Net	56,125.0	42	2.048	88.975	7	0.935
Bayes Net	54,210.0	44	2.046	87.637	8	0.932

Table 6: Results of the three models (training-set view)

Observations: C5 is ranked first on every measure shown, and it uses the fewest fields (6), which makes it the simplest of the three. The three models are very close to each other, with accuracy between about 87.6% and 89.9% and AUC between 0.932 and 0.936. All AUC values are above 0.93, which indicates that the models separate Active and Churned customers well on this data. The names Neural Net and Bayes Net are read from the shortened labels in the model list.

8.2 Predictor Importance

The Predictor Importance chart in the Graph tab shows handset as the most important field, with a clearly longer bar than the others. Fields such as dropped_calls, tariff and age follow with shorter bars, and gender is shown as the least important field. This means that the type of handset a customer uses helps the models most in telling who will churn.

A simple check of the Excel file supports this. The share of customers who churned differs greatly by handset: for ASAD90, CAS30, SOP10 and SOP20 about 95% of customers churned, while for ASAD170, CAS60 and WC95 only about 4% to 6% churned. By comparison, the churn share is almost the same across the four regions (about 45% to 47%). These percentages come from counting the Excel data and are not model outputs.

8.3 Distribution Chart

The Distribution chart in the Graph tab shows the actual churn groups (Active and Churned) coloured by the predicted value. In the Active group, most of the bar is the predicted-Active colour; in the Churned group, most of the bar is the predicted-Churned colour. This shows that the predicted values agree with the actual values for most customers.

8.4 Predictions on the Training Data

The Table node after the nugget shows the original fields along with $XF-churn and $XFC-churn for all 31,781 records. In the 20 rows that were viewed, the confidence values lie between about 0.68 and 0.97, and most predicted values match the actual churn status. A few rows show a different predicted value from the actual one, which is normal because no model is 100% accurate.

8.5 Predictions for the 20 New Customers

The test file has 20 customers. The model predicted 14 as Churned and 6 as Active. The table below lists the predictions exactly as they appear in the output (rows in the same order as the file).

No.	Customer ID	Gender	Age	Tariff	Handset	Dropped calls	Predicted churn	Confidence
1	K261070	Male	35	Play 100	BS210	0	Active	0.901
2	K405340	Female	17	CAT 200	SOP20	15	Churned	0.968
3	K188450	Female	17	CAT 200	ASAD90	2	Churned	0.946
4	K106250	Female	40	CAT 100	ASAD90	0	Churned	0.945
5	K220480	Female	23	Play 100	BS110	0	Churned	0.723
6	K272540	Male	22	Play 100	BS210	0	Active	0.885
7	K297080	Female	29	CAT 100	CAS30	0	Churned	0.944
8	K383500	Male	32	Play 100	S80	0	Active	0.943
9	K149810	Male	30	CAT 200	ASAD90	2	Churned	0.946
10	K196190	Male	21	CAT 200	ASAD170	6	Active	0.963
11	K231470	Female	21	Play 100	S50	0	Churned	0.941
12	K133830	Female	33	CAT 200	ASAD90	2	Churned	0.958
13	K305150	Female	44	CAT 200	BS110	2	Active	0.833
14	K229560	Female	14	Play 100	S50	1	Churned	0.954

No.	Customer ID	Gender	Age	Tariff	Handset	Dropped calls	Predicted churn	Confidence
15	K308740	Female	22	CAT 200	CAS30	1	Churned	0.943
16	K385820	Female	49	Play 100	WC95	1	Active	0.919
17	K185560	Male	21	CAT 200	ASAD90	0	Churned	0.946
18	K189010	Female	24	CAT 200	S50	0	Churned	0.711
19	K335990	Female	22	CAT 50	CAS30	7	Churned	0.954
20	K201530	Male	20	CAT 200	ASAD90	2	Churned	0.947

Table 7: Predicted churn for the 20 test customers

Observations: The model is most confident (0.94 and above) for many customers using handsets ASAD90, SOP20 and CAS30, and least confident for customers 5 and 18 (0.723 and 0.711). Customers with low confidence are the cases where staff should be careful before taking an action.

8.6 Results That Are Not Available

To keep the report honest, the following items cannot be reported because they are not shown in the stream or its outputs:

- Accuracy of the models on a separate test partition, because no Partition node or Analysis node output is available.
- Confusion matrix, precision, recall and F1-score, because no Analysis or Matrix node was used.
- Actual versus predicted comparison for the 20 test customers, because their real churn status is not given.
- The exact list of all 12 models tried by the Auto Classifier, because only the best three are visible in the results.

9. Conclusion, Limitations and Future Scope

9.1 Conclusion

In this practical, a complete churn-prediction stream was built in IBM SPSS Modeler using a telecom dataset of 31,781 customers. The data was read from Excel, field types were defined with the Type node, and the records were sorted by connection date. The Auto Classifier node tried an estimated 12 models and kept three: C5, Neural Net and Bayes Net. In the training-set view, C5 was the best with 89.903% overall accuracy and an AUC of 0.936. Handset was found to be the most important predictor. The combined model was applied to 20 new customers and predicted 14 as Churned and 6 as Active, each with a confidence value.

The practical shows that a business student can build and read a predictive model without coding. For a telecom company, such predictions can help to plan retention offers, handset and tariff strategies, and customer-care calls. The results should be seen as a learning exercise, because the available outputs do not include a separate test evaluation.

9.2 Limitations

- Only the training-set view of results is available, and no separate hold-out test result is shown.
- The data is old (connection dates from 2003 to 2006), so it may not match present customer behaviour.
- Data cleaning was not done in the stream: gender spellings, repeated records and unusual age values are still present.
- The end_date field is almost the same as churn. If a model used it, the result would look better than it really is, so it should be checked or removed.

- The test file has only 20 records, which is too small to judge model quality.

9.3 Future Scope

- Add a Partition node (for example 70% training and 30% testing) and an Analysis node to get a confusion matrix, precision, recall and F1-score.
- Clean the data by merging gender spellings, removing repeated records and handling invalid ages.
- Create new fields, such as how long a customer stayed, and remove fields that carry no information.
- Collect recent data and more fields (for example monthly bill, complaints and data usage).
- Build a dashboard so the marketing team can see the customers most likely to leave.

10. Learning Outcomes

- I learned how to build a stream in IBM SPSS Modeler by joining source, field, record, modeling and output nodes.
- I understood why the Type node is important and how to set a field as Target or Input.
- I learned to read data using Table nodes and to understand $null$ values.
- I learned how the Auto Classifier node compares many models and how accuracy, lift, profit and AUC are used to rank them.
- I learned to read the Predictor Importance chart and to explain which customer details matter most.
- I learned how a generated model nugget is applied to new data and how to read predicted values and confidence.
- I learned to check data quality, such as spelling differences, repeated records, odd ages and fields that leak the answer.
- I understood how churn prediction supports business decisions in marketing and customer retention.
- I learned to report only the results that are actually available and to state clearly what is missing.

11. References

- IBM SPSS Modeler documentation: Auto Classifier node, Type node, Sort node and Table node (IBM Documentation).
- Datasets: telco x customer data.xlsx and telco_test_dataset.xlsx, provided for the practical.
- Project stream file: Stream fp.str (saved from IBM SPSS Modeler).
- College guideline: Predictive Analytics Project Report Format.